Source A · NHS public information

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Cystic fibrosis is inherited. Altered gene copies from both parents cause thick, sticky mucus. This can affect the lungs and digestion. Treatments can help ease symptoms. Care depends on the person.

This source gives a public overview. It does not tell us one person’s exact symptoms or future.

Publisher: NHS. Title: Cystic fibrosis. Use the Causes, Symptoms and Treatments sections.

Source A full page: ask an adult to use the link in the page tools or staff notes.

Accessed 24 October 2026. Page reviewed 28 February 2025. This is a classroom paraphrase, not a quotation. The full page can be opened by the teacher.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Source B · NHS Genomics Education

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Cystic fibrosis usually involves harmful variants in both copies of CFTR. This is an autosomal recessive condition. A carrier has one changed copy in this model. The effects can vary.

Two carrier parents: each pregnancy	Model chance
Child has the condition	1 in 4 = 25%
Child is a carrier	2 in 4 = 50%
Child inherits neither changed copy	1 in 4 = 25%

These chances apply each time. They do not predict a set family count or the severity of one person’s symptoms.

NHS Genomics Education Programme. Page: Cystic fibrosis. Read Genomics, then Inheritance and genomic counselling. Use these parts to check the gene and how changed copies pass to a child.

Source B full page: ask an adult to use the link in the page tools or staff notes.

Checked 24 October 2026. This card uses our own words. It is not a quote. We have not given a review date for this source.

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Sources · purpose and limits

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Both sources are NHS sources. They have different readers and are not wholly independent publishers.

Source A	Source B
Public information about effects and care	More detail about genes and inheritance

Both support inherited cause. More detail does not by itself mean a conflict. Check the actual claim, source section, date and purpose.

Read the source, cover it, explain the meaning in your own words, then check it. Keep the source label beside each claim.

Use fictional cases. Do not ask for family details or predict a person’s future. Inheritance and infection precautions are separate matters.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Supported

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Focus

What are the two main parts of the research question?

Help: Read or point first. Inheritance and a biological effect.

2. Source

How can your audience check a claim?

Help: Read or point first. Make the source visible.

3. Uncertainty

What can you say when the sources do not answer a question?

Help: Read or point first. An honest gap is allowed.

4. Choice

Must this be a public speech?

Help: Read or point first. You may show your work to one person or write it down.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Supported

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Create

Complete the four-panel planning sheet.

Help: Use the four-section outline: question / biological explanation / inheritance model / uncertainty and sources.

2. Present

Communicate the explanation using your chosen mode.

Help: Choose speaking, pointing to a poster or exact-word scribing. Follow the four headings rather than memorising a script.

3. Respond

Answer: “Will every person with CF have exactly the same symptoms?”

Help: Use: the source supports ___, but it does not establish ___. A responsible next check would be ___.

My response: _____

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Use one route for the full 10 minutes. Keep your own record and response.

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Supported

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

1. Explain

Use two sentences. Explain how CF is passed on through genes. Link this to one effect on the body.

Help: Use: my answer is ___ because ___. Point to a model or card first. Use CFTR, two altered alleles and thick mucus.

2. Qualify

State one thing that your sources do not show. Keep this limit in your talk or work.

Help: Use: the evidence shows ___, but cannot yet show ___. The source does not show one person’s future. Say what it does show.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Focus

What are the two main parts of the research question?

Help: Inheritance and a biological effect.

2. Source

How can your audience check a claim?

Help: Make the source visible.

3. Uncertainty

What can you say when the sources do not answer a question?

Help: An honest gap is allowed.

4. Choice

Must this be a public speech?

Help: You may show your work to one person or write it down.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Create

Complete the four-panel planning sheet.

Help: Use one accurate claim per panel rather than long paragraphs.

2. Present

Communicate the explanation using your chosen mode.

Help: A private one-to-one explanation is equally valid.

3. Respond

Answer: “Will every person with CF have exactly the same symptoms?”

Help: Refer to variation and the limits of the sources.

My response: _____

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Use one route for the full 10 minutes. Keep your own record and response.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

1. Explain

Use two sentences. Explain how CF is passed on through genes. Link this to one effect on the body.

Help: Use CFTR, two altered alleles and thick mucus.

2. Qualify

State one thing that your sources do not show. Keep this limit in your talk or work.

Help: The source does not show one person’s future. Say what it does show.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Aspire

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Focus

What are the two main parts of the research question?

Help: Inheritance and a biological effect.

2. Source

How can your audience check a claim?

Help: Make the source visible.

3. Uncertainty

What can you say when the sources do not answer a question?

Help: An honest gap is allowed.

4. Choice

Must this be a public speech? Give a reason.

Help: You may show your work to one person or write it down.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Aspire

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Create

Complete the four-panel planning sheet.

Help: Use one accurate claim per panel rather than long paragraphs.

2. Present

Communicate the explanation using your chosen mode.

Help: A private one-to-one explanation is equally valid.

3. Refine for audience

Choose one sentence with science terms. Make it clear for a new reader. Keep the same meaning and its limits.

Help: Keep the source name. State what it does not tell us about one person.

My response: _____

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Use one route for the full 10 minutes. Keep your own record and response.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Aspire

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

1. Explain

Use two sentences. Explain how CF is passed on through genes. Link this to one effect on the body.

Help: Use CFTR, two altered alleles and thick mucus.

2. Refine for audience

Choose one sentence with science terms. Make it clear for a new reader. Keep the same meaning and its limits.

Help: Keep the source name. State what it does not tell us about one person.

My response: _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Word page

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Word	Meaning
Source	Where the information comes from.
Publisher	Who is responsible for the source.
Paraphrase	Explain the meaning in your own words.
Carrier	One altered allele in this recessive model.
Autosomal recessive	Two altered copies are usually needed. The gene is on a non-sex chromosome.
Uncertainty	What the source does not establish.

Say the idea in your own words. Keep its meaning and name the source.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Knowledge organiser

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Present an accurate, source-linked explanation and respond to a question without exceeding the evidence.

Show how CF passes through genes. Link this to an effect on the body. Name the source for each claim. Hear a question. Say what your sources show and what they do not show.

Research step	What I do
Find	Keep a bounded question and two named sources.
Label	Keep publisher, title, date, purpose and claim together.
Map	Link inheritance to one effect on the body.
Gap	State what the sources do not establish.

CF usually involves altered CFTR alleles from both parents. Thick mucus can affect lungs and digestion. A model chance does not predict one person’s exact symptoms.

Source A: NHS public information. Source B: NHS Genomics Education. Their different emphasis need not be a conflict.

Use fictional cases. Do not ask for family details or predict a person’s future. Inheritance and infection precautions are separate matters.

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Starter

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Which answer to an audience question is strongest?

A. I will make up a confident answer

B. That is outside what my sources establish; I can identify what evidence would be needed

C. Everyone with the condition is exactly the same

Keep a first choice. Give the clue before you check the answer.

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

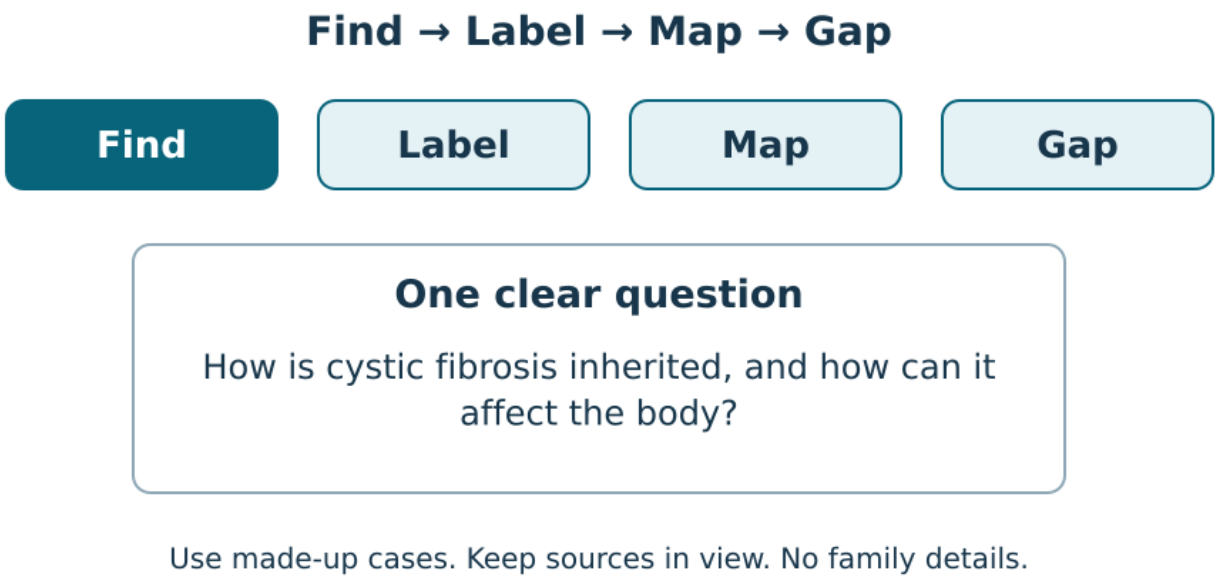
Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Find

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.



Use a clear question. Find two NHS sources. Keep the question beside your work. What does each source help you find out?

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Label

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.



Name who wrote it and the page title. Keep the date and the claim. Show the line you used. What does it support?

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

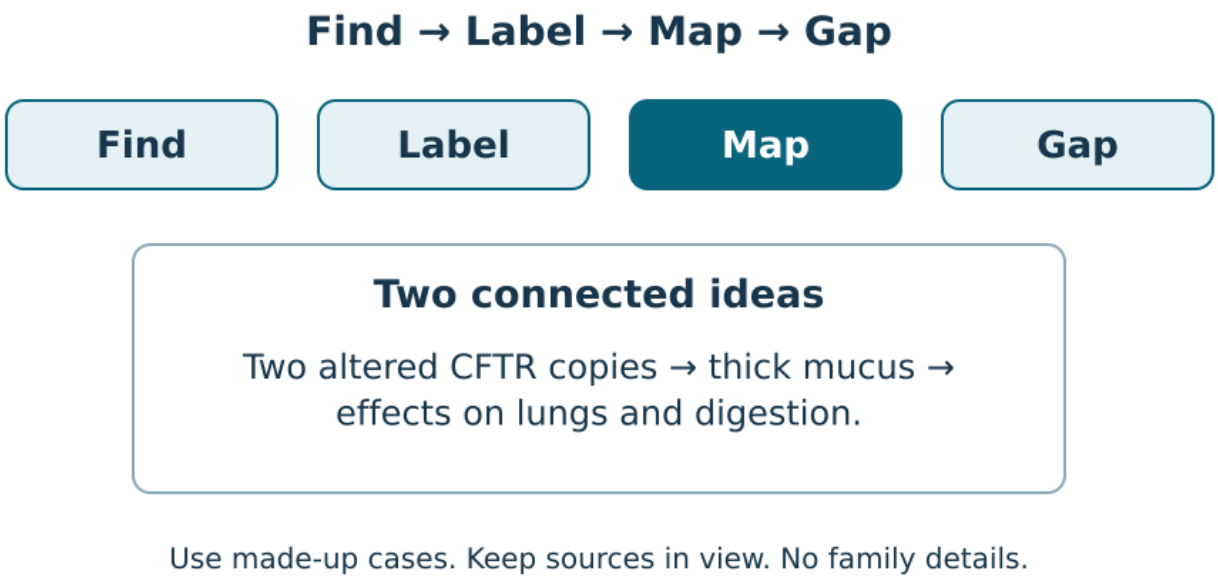
Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Map

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.



Link the gene copies to an effect on the body. Keep one limit. Point to each link and say what the sources show.

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

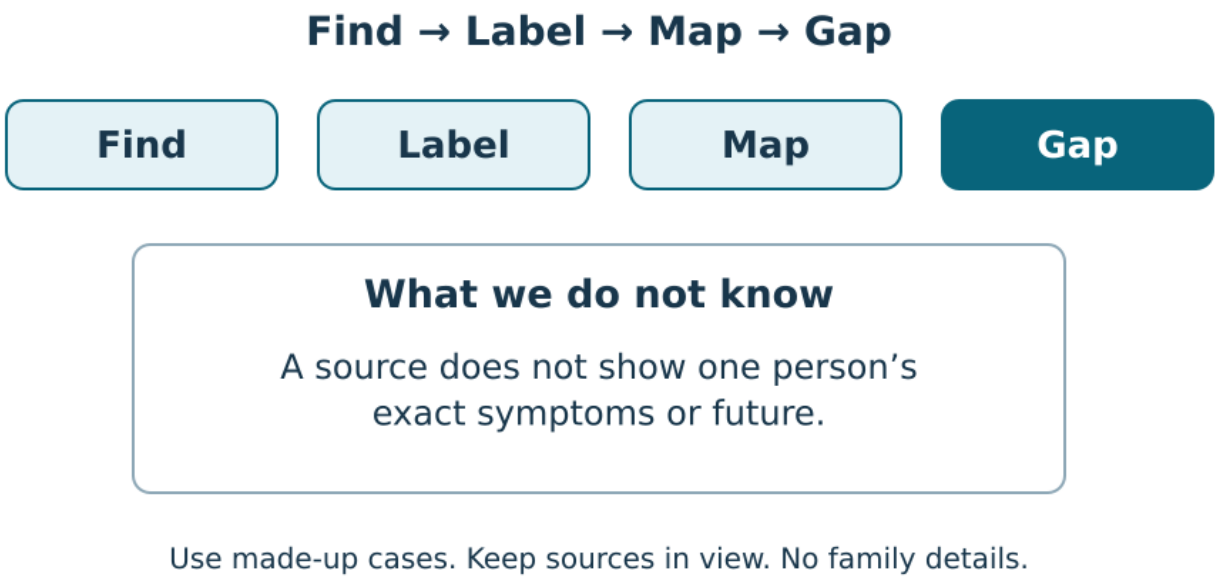
Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Keep a gap visible

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.



Use made-up cases. Keep sources in view. No family details.

Say what the source shows. Then say what it cannot show about one person. What would you need to check next?

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · sort with evidence

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Choose a group for each card. Give the source clue for one choice.

Card	Your group
A bounded question	_____
An accurate inheritance model	_____
Visible source labels	_____
A stated uncertainty	_____
Private details about a classmate	_____
A guaranteed prediction for one person	_____

Groups: Keep in my work / Remove or change.

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

I do 2 · a worked example

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Hear the question. Which part can your sources answer?

Point to the line that helps. Say what it shows.

If there is a gap, say so. The source may show how CF is passed on but not one person’s future.

Name the evidence you would need. Do not make up an answer to fill the gap.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Four panels · planning sheet

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Plan one claim for each panel. Show the source for each claim. Use two named sources. Keep one limit. You may show it to your partner, write or direct a scribe.

Panel	My own explanation
1. A bounded question	_____
2. Recessive inheritance model	_____
3. One effect and its source	_____
4. Two sources and one uncertainty	_____

My chosen mode: _____

Made from the web lesson SCI-LAUNCH-A2-W06-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Audience question and my response

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Audience question: will every person with CF have exactly the same symptoms?

What my sources support: _____

What they do not establish: _____

A responsible next check: _____

What the adult heard: _____

My next response, or why I kept it: _____

Keep your first answer. Hear what the adult noticed. Make one real change, or give a reason to keep it. Ask for a read-back if it helps.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale